

# ARSHATH HAMEED K

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Portfolio: [arshath4758u.github.io/arshath-portfolio](https://arshath4758u.github.io/arshath-portfolio)

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## SUMMARY

AI/ML and LLM post-training intern with hands-on experience at Vdart and Ethara AI, spanning multi-agent systems, RLHF pipelines, LLM post-training, and production-grade model evaluation. Proficient with DeepSeek, Gemini, Claude, and Llama; experienced across the full LLM development lifecycle from dataset curation and prompt engineering to alignment and reward modeling. Co-inventor of a filed patent in multi-agent reinforcement learning for AI-driven data center optimization.

## SKILLS

**AI & LLMs:** DeepSeek, Gemini, Claude, LLM Fine-Tuning, Ollama, Multi-Agent Systems, RAG, Prompt Engineering, RLHF, Model Evaluation, Machine Learning, Deep Learning, Hugging Face, Computer Vision, NLP, Object Detection, Document Parsing

**Tech Stack:** PyTorch, TensorFlow, Transformers, LangChain, YOLOv8, FastAPI, Flask, Python, MongoDB, AWS, Google Cloud

**Languages & Tools:** JavaScript, GitHub, PowerBI, Excel

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## EXPERIENCE

### Artificial Intelligence & Machine Learning Intern | Vdart | Tiruchirappalli

June 2025 – January 2026

#### Orca-AI: Generative AI Application Generator

- Architected multi-agent AI system that generates full-stack web applications from natural language prompts using Llama 3 / Phi-3 with Judge AI agent for autonomous code refinement, validation, and quality assurance; implemented real-time WebSocket collaboration for seamless multi-user development
- **Tech:** Ollama, LangChain, Llama 3, Phi-3, FastAPI, GitHub API, WebSockets

#### Smart Interview System: Computer Vision & Proctoring

- Engineered comprehensive AI-powered interview platform with YOLOv8-based real-time violation detection, SpeechBrain audio analysis for voice authentication and speech pattern recognition, and WebRTC video streaming with frame-level analytics for instant proctoring dashboard and candidate scoring
- **Tech:** YOLOv8, OpenCV, PyTorch, SpeechBrain, WebRTC, FastAPI

#### AI-Powered Resume Parser: NLP Recruitment Portal

- Developed end-to-end NLP pipeline with Transformers-based named entity recognition (NER) for automated resume parsing and semantic candidate matching; engineered bulk document processing with Google Drive integration for enterprise-scale recruitment automation
  - **Tech:** Transformers, spaCy, Flask, MongoDB, Google Drive API, Semantic Search
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## LLM Post-Trainee Intern | Ethara AI | Gurugram, Haryana (Remote)

February 2026 – May 2026

- Executed end-to-end RLHF workflows on the Multimango platform — covering data annotation, preference ranking, and reward signal generation — to improve LLM output quality, instruction-following accuracy, and alignment across diverse task categories.
- Conducted structured model evaluation across multimodal outputs (text, visual, and structured data), identifying hallucinations, reasoning failures, and formatting inconsistencies to provide high-fidelity feedback signals for reward model training.
- Engineered adversarial and constraint-aware prompts to stress-test model behavior, uncovering edge cases in instruction-following and reasoning; curated and corrected training datasets to meet precision standards required for post-training pipelines.
- Contributed calibrated quality assessments on model-generated technical diagrams, spatial layouts, and localized data outputs, directly supporting iterative reward model improvements across multimodal domains.
- **Tech:** Multimango, RLHF, Reward Modeling, Data Annotation, Model Evaluation

## PATENT FILING & INNOVATION

**"JANUS: System and Method for Holistic, AI-Driven Thermal and Energy Optimization in Data Centers via Multi-Agent Reinforcement Learning"** | Filed December 2025

- Co-invented a novel 7-module AI architecture validated via simulation to outperform single-agent cooling systems by 15–20 percentage points in energy efficiency.
- Designed the core AI architecture combining **Physics-Informed Graph Neural Networks, Transformers, and Multi-Agent Reinforcement Learning (MARL)** to predict and prevent thermal spikes up to 48 hours in advance.
- Built a **Causal AI module** that autonomously identifies root causes of thermal faults and runs counterfactual simulations — eliminating the need for real-world failure testing.
- Integrated a **financial optimization layer** that schedules workloads based on real-time electricity pricing and SLO priority, reducing operational costs through intelligent load shifting

**Status:** Provisional Specification filed under Patents Act, 1970.

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## EDUCATION

**Bachelor of Engineering (B.E.) – Computer Science & Engineering** | K. Ramakrishnan College of Engineering, Tiruchirappalli | **CGPA: 8.1/10** | 2021 – 2025

**Higher Secondary (+2) & Secondary (SSLC)** | **HSC: 88.2%** | **SSLC: 85%** | 2018 – 2021

## CERTIFICATIONS

AWS Certified Cloud Practitioner (2024) | Google Cloud Associate Cloud Engineer (2024) | Salesforce Administrator (2024) | NPTEL - Human Computer Interaction and Java